

Bernoulli
 $X \sim \text{Bernoulli}(p), \quad 0 \leq p \leq 1$
 $f_X(x) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\}$
 $\mathbb{E}[X] = p \qquad \text{Var}(X) = p(1-p)$
Binomial
 $X \sim \text{Binomial}(n, p), \quad 0 \leq p \leq 1$

$X \stackrel{d}{=} Z_1 + \cdots + Z_n, \quad Z_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$
 $f_X(x) = \binom{n}{x} p^x(1-p)^{n-x}, x \in \{0\} \cup \mathbb{N}$
 $\mathbb{E}[X] = np \qquad \text{Var}(X) = np(1-p)$
 $M_X(t) = (pe^t + q)^n, \quad q = 1-p$
Multinomial
 $\mathbf{X} \sim \text{Multi}(n, (p_1, \cdots, p_k)^t)$
 $p_j \geq 0, \sum p_j = 1, \sum x_i = n$
 $f_{\mathbf{X}}(\mathbf{x}) = \binom{n}{x_1 \cdots x_k} p_1^{x_1} \cdots p_k^{x_k}$
 $M_X(t) = (p_1 e^{t_1} + \cdots p_k e^{t_k})^n$
Geometric
 $X \sim \text{Geo}(p), \quad 0 < p < 1$
 $\text{Geo}(p) = \text{Negbin}(1, p)$
 $f_X(x) = p(1-p)^{x-1}, \quad x \in \{1, 2, \dots\}$
 $\mathbb{E}[X] = 1/p \qquad \text{Var}(X) = (1-p)/p^2$
Negative Binomial
 $X \sim \text{Negbin}(r, p), \quad 0 < p < 1, r \in \mathbb{N}$

$X \stackrel{d}{=} Z_1 + \cdots + Z_r, \quad Z_i \stackrel{iid}{\sim} \text{Geo}(p)$
 $f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, x \geq r, x \in \mathbb{N}$
 $\mathbb{E}[X] = r/p \qquad \text{Var}(X) = r(1-p)/p^2$
 $M_X(t) = (pe^t(1-qe^t)^{-1})^r, \quad t < -\log q$
Poisson
 $X \sim \text{Poisson}(\lambda), \quad \lambda \geq 0$
 $\text{Poisson}(\lambda) = \lim_{n \rightarrow \infty, np \rightarrow \lambda} \text{Binomial}(n, p)$
 $f_X(x) = e^{-\lambda} \lambda^x / x!, \quad x \in \{0\} \cup \mathbb{N}$
 $\mathbb{E}[X] = \lambda, \text{Var}(X) = \lambda$
 $M_X(t) = e^{-\lambda + \lambda e^t}$
Exponential
 $X \sim \text{Exp}(1/\lambda), \quad \lambda > 0$
 $\text{Exp}(1/\lambda) \stackrel{d}{=} \text{Gamma}(1, 1/\lambda)$

$f_X(x) = \lambda e^{-\lambda x}, \quad x > 0$
 $F_X(x) = 1 - e^{-\lambda x}, \quad x > 0$
 $F_X^{-1}(x) = -1/\lambda \log(1-x), \quad 0 < x < 1$
 $\mathbb{E}[X] = 1/\lambda \qquad \text{Var}(X) = 1/\lambda^2$
 $M_X(t) = \lambda/(\lambda - t), \quad t < \lambda$
Gamma
 $X \sim \text{Gamma}(\alpha, \beta), \quad \alpha, \beta > 0$

$\alpha \in \mathbb{N}, X \stackrel{d}{=} Z_1 + \cdots + Z_{\alpha}, \quad Z_i \stackrel{iid}{\sim} \text{Exp}(\beta)$
 $X_i \stackrel{iid}{\sim} \text{Gamma}(\alpha_i, \beta)$
 $\sum X_i \sim \text{Gamma}\left(\sum \alpha_i, \beta\right)$
 $f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x > 0$
 $\mathbb{E}[X] = \alpha\beta \qquad \text{Var}(X) = \alpha\beta^2$

$M_X(t) = (1 - \beta t)^{-\alpha}, \quad t < 1/\beta$
Gamma Function
 $\int_0^\infty x^{\alpha-1} e^{-x} dx \triangleq \Gamma(\alpha), \quad \alpha > 0$
 $\Gamma(x+1) = x\Gamma(x), \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \Gamma(1) = 1$
Normal
 $X \sim \mathcal{N}(\mu, \sigma^2), \quad \sigma > 0$
 $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$
 $\mathbb{E}[X] = \mu \qquad \text{Var}(X) = \sigma^2$
 $M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$

Dirichlet
 $Y_1, \dots, Y_k \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_{k+1})$
 $X_1, \dots, X_{k+1} \stackrel{iid}{\sim} \text{Gamma}(\alpha_i, \beta), \quad \alpha_i > 0$
 $Y_i = \frac{X_i}{X_1 + \cdots + X_{k+1}}, \quad i = 1, \dots, k$
 $f_Y(y_1, \dots, y_k)$
 $= \frac{\Gamma(\alpha_1 + \cdots + \alpha_{k+1})}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_{k+1})} \prod_{i=1}^k y_i^{\alpha_i-1}$
 $(1-y_1 \cdots -y_k)^{\alpha_{k+1}-1}$
 $I_{(y_i > 0, y_1 + \cdots + y_k < 1)}$
 $k=1, \quad Y_1 \sim \text{Beta}(\alpha_1, \alpha_2)$
Chi-square
 $X_1^2 + \cdots + X_r^2 \sim \chi^2(r), X_i \stackrel{iid}{\sim} \mathcal{N}(0, 1)$

$\chi^2(r) = \text{Gamma}\left(\frac{r}{2}, 2\right)$
Student's t
 $X = \frac{Z}{\sqrt{V/r}} \sim t(r), \quad r > 0$
 $Z \sim \mathcal{N}(0, 1), V \sim \chi^2(r), Z \perp V$
 $t(1) \stackrel{d}{=} \text{Cauchy}(0, 1)$
Cauchy
 $X \sim \text{Cauchy}(\mu, \sigma), \quad \sigma > 0$
 $= \mu + \sigma Z, \quad Z \sim \text{Cauchy}(0, 1) \stackrel{d}{=} t(1)$

$f_X(x) = \frac{1}{\pi\sigma} \left(1 + \left(\frac{x-\mu}{\sigma}\right)^2\right)^{-1}$
F Distribution
 $X \sim \text{F}(r_1, r_2) = \frac{V_1/r_1}{V_2/r_2}$
 $V_1 \sim \chi^2(r_1), V_2 \sim \chi^2(r_2), V_1 \perp V_2$
 $F^{-1} \sim \text{F}(r_2, r_1), \quad t^2(r) \sim \text{F}(1, r)$
Thm of Normal Sampling
 $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$
 $(i) \overline{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
 $(ii) s^2 = \frac{1}{n-1} \sum (X_i - \overline{X})^2 \perp \overline{X}$
 $(iii) \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$
Probability Integral Transformation
 F is strictly increase

$F(X) \sim \text{U}(0, 1), F^{-1}(U) \stackrel{d}{=} X$
 $U_{(r)} \stackrel{d}{=} 1 - e^{-Z_{(r)}}, \quad Z_r \sim \text{Exp}(1)$
Uniform Order Statistics
 $X_{(1)}, \dots, X_{(n)} \sim \text{U}(0, 1)$
 $X_{(r)} \sim \text{Beta}(r, n-r+1)$
 $(X_{(1)}, X_{(2)} - X_{(1)}, \dots, X_{(n)} - X_{(n-1)})$
 $\sim \text{Dirichlet}(1, \dots, 1)$
Exponential Order Statistics
 $X_{(1)}, \dots, X_{(n)} \sim \text{Exp}(1), \quad Z_r \stackrel{iid}{\sim} \text{Exp}(1)$
 $X_{(r)} \stackrel{d}{=} \left(\frac{1}{n} Z_1 + \cdots + \frac{1}{n-r+1} Z_r\right)$
Definition of Order Statistics
 $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} F(x)$
 $h(y) = F^{-1}(1 - e^{-y})$
 $(X_{(r)}) \stackrel{d}{=} (F^{-1}(U_{(r)}))$
 $\stackrel{d}{=} (F^{-1}(1 - e^{-Z_{(r)}})) \stackrel{d}{=} h(Z_{(r)})$
Multivariate Normal Distribution
 $\mathbf{X} \stackrel{d}{=} \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2} \mathbf{Z}$
 $f_{\mathbf{X}}(\mathbf{x}) = |2\pi \boldsymbol{\Sigma}|^{-1/2}$
 $\exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$

$m_{\mathbf{X}}(\mathbf{t}) = \exp(\boldsymbol{\mu}^T \mathbf{t} + \frac{1}{2} \mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t})$
Convergence in Distribution
 $\forall x: x \in \text{Conti}(F_Z)$
 $\lim_{n \rightarrow \infty} P(X_n \leq x) = P(Z \leq x)$
 $\forall \mathbf{x}: \mathbf{x} \in \text{Conti}(F_Z)$
 $\lim_{n \rightarrow \infty} P\left(\bigcap_i \mathbf{X}_{ni} \leq \mathbf{x}_i\right) = P\left(\bigcap_i \mathbf{Z}_i \leq \mathbf{x}_i\right)$
Central Limit Theorem
 $X_1, \dots, X_n \stackrel{iid}{\sim} (\mu, \sigma^2), \text{Var}(X_1) < \infty$

$\sqrt{n}(\overline{X_n} - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$
 $\mathbf{X}_1, \dots, \mathbf{X}_n \stackrel{iid}{\sim} (\boldsymbol{\mu}, \boldsymbol{\Sigma}), \text{Var}(\mathbf{X}_1) < \infty$
 $\sqrt{n}(\overline{\mathbf{X}_n} - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}_k(\mathbf{0}, \boldsymbol{\Sigma})$
Convergence in Probability
 $\lim_{n \rightarrow \infty} P(|X_n - c| > \epsilon) = 0 \quad \forall \epsilon > 0$
 $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$
 $\iff \lim_{n \rightarrow \infty} P(|\| \mathbf{X}_n - \mathbf{c} \| > \epsilon) = 0 \forall \epsilon > 0$
 $\iff \lim_{n \rightarrow \infty} P(\max_i |\mathbf{X}_{ni} - c_i| > \epsilon) = 0$
 $\iff \mathbf{X}_{nj} \xrightarrow{p} c_j \quad \forall j$
Law of Large Numbers
 $X_1, \dots, X_n \stackrel{iid}{\sim} \mathbb{E}(X_1) < \infty, \overline{X_n} \xrightarrow{p} \mathbb{E}(X_1)$

Slutsky Theorem
 $X_n \xrightarrow[n \rightarrow \infty]{d} Z, Y_n \xrightarrow{p} c$
 $+, -, *, /$ are well defined
Delta Method
 $\sqrt{n}(\mathbf{X}_n - \boldsymbol{\theta}) \xrightarrow[n \rightarrow \infty]{d} Z, \dot{g}(\boldsymbol{\theta})$ is Continuous
 $\sqrt{n}(g(\mathbf{X}_n) - g(\boldsymbol{\theta})) \xrightarrow[n \rightarrow \infty]{d} (\dot{g}(\boldsymbol{\theta}))^T Z$

Regularity Condition
(R0) $f(x; \theta) = f(x; \theta^0) \Rightarrow \theta = \theta^0$
(R1) $\{x: f(x; \theta) > 0\}$ 는 θ 와 무관
(R2) Ω 는 k차원 열린 집합
(R3) $\forall x, \dot{l}_n(\theta), \ddot{l}_n(\theta)$ 존재, 연속
(R4) $\partial_\theta \mathbb{E}[u(X)] = \int_{\mathbb{R}^n} u(x) \partial_\theta (f(x; \theta)) dx$
(R5) $\exists [I(\theta)]^{-1} = \text{Var}_\theta(\partial_\theta \log f(X_1; \theta))^{-1}$
(R6) $\exists! \hat{l}_n^{\text{MLE}} \quad \dot{l}_n(\hat{\theta}) = 0, \quad \hat{\theta}_n \xrightarrow{p} \theta$
(R7) $\exists E(M(X_1)) < \infty$, 삼차도함수 연속
 $\max_{\theta \in \Omega} \|\partial_\theta^3 \log f(X_1; \theta)\| \leq M(X_1)$

Exponential Family
 $f(x; \boldsymbol{\eta}) = \exp\left(\sum_{j=1}^k \eta_j T_j(x) - A(\boldsymbol{\eta}) + S(x)\right)$
(R1), (R2), $c^t T(x)$ 는 상수가 아닐 때
 $\mathbb{E}_{\boldsymbol{\eta}}(T(X)) = \dot{A}(\boldsymbol{\eta}), \text{Var}(T(X)) = \ddot{A}(\boldsymbol{\eta})$

$\dot{A}(\boldsymbol{\eta}) = \mathbb{E}_{\boldsymbol{\eta}}(T(X_1)) = \frac{1}{n} \sum_{i=1}^n T(x_i)$
Kullback-Leibler divergence
 $\text{KL}(\theta, \theta^0) = -\mathbb{E}_{\theta^0}[\log\{f(X; \theta)/f(X; \theta^0)\}]$
(R0), (R1)일 때
(i) $\text{KL}(\theta, \theta^0) \geq 0 \quad \forall \theta \in \Omega$
(ii) $\text{KL}(\theta, \theta^0) = 0 \iff \theta = \theta^0$
argmax consistency (R6)
 $\overline{l}_n(\theta) \xrightarrow{p} \mathbb{E}_{\theta^0}[\log f(X_1; \theta)] \quad (\text{LLN})$
 $\begin{cases} \text{(R0), (R1)} \\ c^t \dot{l}_n(\theta) c < 0, \forall c \neq 0, \forall \theta \in \Omega \\ \lim_{\theta \rightarrow \partial(\Omega)} l_n(\theta) = -\infty \\ \forall \theta^0 \in \Omega, \mathbb{E}_{\theta^0}[\log f(X_1; \theta)] \text{ 존재, } \theta \in \Omega \text{ 연속} \end{cases}$
 $\rightarrow \text{(R6)}$

Score Function
(R0)-(R5)일 때
(a) $\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X_1; \theta) \right] = 0$
(b) $I(\theta) = \text{Var}_\theta \left[\frac{\partial}{\partial \theta} \log f(X_1; \theta) \right]$
 $= \mathbb{E}_\theta \left[-\frac{\partial^2}{\partial \theta^2} \log f(X_1; \theta) \right]$

Asymptotic Normality of MLE
(R0)-(R7) $\Rightarrow \sqrt{n}(\hat{\theta}_n^{\text{MLE}} - \theta) \xrightarrow{d} \mathcal{N}(0, I(\theta)^{-1})$
One-step newton-Raphson estimator
(R0)-(R7)일 때

$\sqrt{n}(\hat{\theta}_n^{(0)} - \theta) \xrightarrow{d} Z, \hat{\theta}_n^{(1)} = \hat{\theta}_n^{(0)} - \frac{\tilde{l}_n(\hat{\theta}_n^{(0)})}{\ddot{l}_n(\hat{\theta}_n^{(0)})}$
 $\sqrt{n}(\hat{\theta}_n^{(1)} - \theta) \xrightarrow[n \rightarrow \infty]{d} \mathcal{N}(0, I(\theta)^{-1})$
Cramer-Rao Lower Bound
(R0)-(R5), $\eta \in \mathbb{R}, \forall \theta \in \Omega$,
 $\text{Var}_\theta(\hat{\eta}_n)$
 $\geq \left(\frac{\partial}{\partial \theta} \mathbb{E}_\theta(\hat{\eta}_n)\right)^t [nI(\theta)]^{-1} \left(\frac{\partial}{\partial \theta} \mathbb{E}_\theta(\hat{\eta}_n)\right)$
Likelihood Ratio Test
 $H_0: \theta \in \Omega_0, H_1: \theta \in \Omega \setminus \Omega_0 = \Omega_1$

$2\log \Lambda = 2(l(\hat{\theta}^\Omega; x) - l(\hat{\theta}^{\Omega_0}; x))$
Reject Region
 $C_\alpha = \{x: 2(l(\hat{\theta}^\Omega; x) - l(\hat{\theta}^{\Omega_0}; x)) \geq c\}$
 $\max_{\theta \in \Omega_0} \gamma(\theta) = \max_{\theta \in \Omega_0} P_\theta\{(X_1, \dots, X_n) \in C_\alpha\}$
 $= \alpha$
Asymp. Dist. of LRT Statistic
 $\theta \in \Omega \subseteq \mathbb{R}^k, \theta = (\xi^t, \eta^t)^t, \quad \text{(R0)-(R7)}$
 $H_0: \xi = \xi_0, \Omega_0 = \{(\xi_0^t, \eta^t)^t \in \Omega: \eta \in \mathbb{R}^{k_0}\}$
(a) $2\{l(\widehat{\theta}_n^\Omega) - l(\widehat{\theta}_n^{\Omega_0})\}$
(b) $n(\widehat{\theta}_n^\Omega - \widehat{\theta}_n^{\Omega_0})^t I(\theta_0)(\widehat{\theta}_n^\Omega - \widehat{\theta}_n^{\Omega_0})$
(c) $n\tilde{l}(\widehat{\theta}_n^{\Omega_0})^t [I(\theta_0)]^{-1} \tilde{l}(\widehat{\theta}_n^{\Omega_0})$
 $\xrightarrow{d} \chi^2(k - k_0)$
 $W_n(\hat{\theta}_n^{\Omega_0}) = n(\hat{\theta}_n - \hat{\theta}_n^{\Omega_0})^t I(\hat{\theta}_n^{\Omega_0})(\hat{\theta}_n^\Omega - \hat{\theta}_n^{\Omega_0})$
 $R_n(\hat{\theta}_n^{\Omega_0}) = n\tilde{l}(\hat{\theta}_n^{\Omega_0})^t [I(\hat{\theta}_n^{\Omega_0})]^{-1} \tilde{l}(\hat{\theta}_n^{\Omega_0})$
Sufficient Statistic
 $Y = u(X_1, \dots, X_n)$ is sufficient for θ

$\Leftrightarrow \prod_{i=1}^n f(x_i; \theta) = k_1(u(x), \theta) k_2(x)$
 $\Leftrightarrow P((X_1, \dots, X_n)^t \in A | Y = y)$ 는 θ 와 무관
 Y 의 일대일 대응인 $W = g(Y)$ 는 충분통계량
완비성: $\mathbb{E}_\theta(g(Y)) = 0 \forall \theta \in \Omega$
 $\Rightarrow P(g(Y) = 0) = 1$

지수족: $\left(\sum_{i=1}^n T_1(X_i), \dots, \sum_{i=1}^n T_k(X_i)\right)$
Ancillary Statistic
 $Z = v(X_1, \dots, X_n)$ is ancillary for θ
 $\Leftrightarrow P(Z = z)$ 는 θ 와 무관
 Y 가 c.s.s.이면, $Y \perp Z$
Rao-Blackwell Theorem
(a) Y 는 θ 의 충분통계량
(b) $\hat{\eta}(X_1, \dots, X_n)$ 는 θ 의 추정량
(c) $\hat{\eta}^{RB}(Y) = \mathbb{E}_\theta[\hat{\eta}(X_1, \dots, X_n) | Y]$
 $\mathbb{E}_\theta(\hat{\eta}^{RB}) = \mathbb{E}_\theta(\hat{\eta}), \text{Var}_\theta(\hat{\eta}^{RB}) \leq \text{Var}_\theta(\hat{\eta})$
Misc.

$a_n = o(b_n) \Leftrightarrow \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$
 $-\log(1-x) = x + \frac{1}{2}x^2 + o(x^2)$
 $\rho_4 = \mathbb{E}\left[\left(\frac{X_1 - \mu}{\sigma}\right)^4\right] - 3$
 $\sqrt{n}(s_n^2 - \sigma^2) \xrightarrow{d} \mathcal{N}(0, (\rho_4 + 2)\sigma^4)$
 $\sqrt{n}(\hat{\rho}_n - \rho)$
 $\xrightarrow{d} \mathcal{N}(0, \text{Var}(X_1 Y_1 - \frac{\rho}{2} X_1^2 - \frac{\rho}{2} Y_1^2))$
 $\frac{1}{2} \sqrt{n} \left(\log \frac{1 + \hat{\rho}_n}{1 - \hat{\rho}_n} - \log \frac{1 + \rho}{1 - \rho} \right) \xrightarrow{d} \mathcal{N}(0, 1)$
 $\begin{cases} \sqrt{n}(k_i/n - q_i) \rightarrow 0, \quad i < j \Rightarrow q_i < q_j \\ \xi_{q_i} = F^{-1}(q_i) \\ \sqrt{n}(\mathbf{X}_{k_i:n} - \boldsymbol{\xi}_{q_i}) \xrightarrow{d} \mathcal{N}(0, \Sigma) \\ \Sigma_{ij} = \frac{q_i(1-q_j)}{f(\xi_{q_i})f(\xi_{q_j})} \end{cases}$
 $(A + uv^t)^{-1} = A^{-1} - \frac{A^{-1} u v^t A^{-1}}{1 + v^t A^{-1} u}$